

Autonomous Computational Framework for Multi-Phase Impact Dynamics and Terminal Ballistics in Reinforced Geomaterials

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Abstract—The physical characterization of high-velocity terminal ballistics against deeply buried, hardened target structures (DBHTs) presents a complex multi-physics challenge encompassing aerodynamic free-fall, dynamic cavity expansion, hydrodynamic rod erosion, and shock-wave material degradation. Conventional analytical calculators often lack the fidelity required to model non-linear material transitions, while enterprise finite-element hydrocodes remain computationally prohibitive for large-scale parametric exploration. This paper introduces the MOP Simulator V3.0, an open-source, full-stack computational research platform that unifies a high-performance C++23 continuum mechanics engine with an automated Node.js orchestration backend and large language model (LLM) scenario synthesizers. The physics kernel couples Runge-Kutta 4th-Order (RK4) numerical integration with the two-phase Forrestal cavity expansion model, the Walker-Anderson hydrodynamic rod erosion model (WAPM), CEB-FIP Dynamic Increase Factors (DIF), and Walker-Wasley Hugoniot shock initiation criteria. The automation subsystem enables headless execution, memory-safe asynchronous telemetry streaming (1,000-frame chunking into MongoDB), and automated statistical aggregation. We demonstrate the platform across extensive multi-cycle simulation campaigns, characterizing penetration regimes, casing survivability envelopes, and sequential multi-bomb salvo synergies against 70 MPa reinforced concrete and monolithic granite overburden. Finally, we establish a theoretical and architectural framework for V4.0 surrogate neural physics integration via LibTorch and deep reinforcement learning smart fuzing.

Index Terms—Terminal Ballistics, Cavity Expansion Theory, Hydrodynamic Erosion, Hugoniot Equation of State, Alekseevskii-Tate Model, Dynamic Increase Factor, Autonomous Scientific Computing, LibTorch, High-Velocity Penetration.

I. INTRODUCTION

DEFEATING deeply buried hardened targets (DBHTs)—such as underground command facilities, reinforced concrete silos, and tunnel networks—requires penetrating munitions capable of sustaining structural integrity through extreme deceleration forces, dynamic shock stresses, and intense thermal friction before detonating at an optimal subterranean depth. The 13,608 kg GBU-57 Massive Ordnance Penetrator (MOP) represents the empirical archetype of conventional bunker-defeat capabilities, relying on a high sectional density, hardened alloy casing, and delayed fuzing systems.

Historically, computational analysis of terminal impact physics has been divided into two separate paradigms:

- 1) **Simplified Semi-Empirical Formulations:** Low-overhead empirical equations (e.g., Young, NDRC, or classic Poncelet models) that predict final penetration depth with modest accuracy but fail to capture transient phenomena such as interface erosion, shock-induced casing failure, or atmospheric trajectory deviation.
- 2) **High-Fidelity Hydrocodes:** Full Eulerian or Arbitrary Lagrangian-Eulerian (ALE) hydrocode simulations (e.g., LS-DYNA, Autodyn, CTH) that model complete stress tensors and material fractures. While physically comprehensive, a single hydrocode impact run can demand hours or days of high-performance computing (HPC) cluster time, making thousands of iterative parametric sweeps computationally intractable.

To bridge this operational gap, we developed the **MOP Simulator**, a modular, high-throughput computational platform designed to perform sub-millisecond, multi-phase impact trajectory calculations while maintaining rigorous adherence to established continuum mechanics and terminal ballistics literature. In V3.0, the platform was re-architected into a full-stack autonomous system capable of executing self-directed parametric sweeps, persisting high-frequency telemetry in a database, and synthesizing formal scientific research papers.

II. MATHEMATICAL & PHYSICAL FORMULATIONS

The simulation engine models the complete physics lifecycle of a penetrating projectile across two distinct phases: the **Atmospheric Free-Fall Drop Phase** and the **Subterranean Ground Penetration Phase**.

A. Atmospheric Free-Fall & Aerodynamics

The projectile is released at an altitude y_0 (typically 40,000–50,000 ft) with an initial release velocity v_0 . Trajectory kinematics are integrated using a 4th-Order Runge-Kutta (RK4) numerical solver with a continuous time-step $\Delta t = 0.01$ s.

Ambient atmospheric conditions are dynamically updated as a function of altitude y using the **US Standard Atmosphere (1976)** model. The barometric density $\rho_{air}(y)$ and speed of sound $c(y)$ are formulated as:

$$\rho_{air}(y) = \rho_0 \cdot \exp\left(-\frac{M_{air} \cdot g \cdot y}{R \cdot T(y)}\right) \quad (1)$$

$$c(y) = \sqrt{\gamma_{air} \cdot \frac{R}{M_{air}} \cdot T(y)} \quad (2)$$

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where $\rho_0 = 1.225 \text{ kg/m}^3$, $M_{air} = 0.0289644 \text{ kg/mol}$, $R = 8.31446 \text{ J/(mol} \cdot \text{K)}$, and $\gamma_{air} = 1.4$.

Aerodynamic drag force F_d opposing the velocity vector $\vec{v}$ is evaluated based on the projectile's cross-sectional area $A = \frac{\pi D^2}{4}$ and nose geometry:

$$F_d = \frac{1}{2} \rho_{air}(y) \cdot v^2 \cdot A \cdot C_d(M) \quad (3)$$

The drag coefficient $C_d(M)$ incorporates both the geometric Caliber-Radius-Head ratio ($\text{CRH} = \frac{R_{nose}}{D}$) and compressibility corrections across subsonic, transonic, and supersonic regimes:

$$C_{d,base} = \frac{8 \cdot \text{CRH} - 1}{24 \cdot \text{CRH}^2} \quad (4)$$

B. Shock Impedance Matching & Walker-Wasley Initiation

Upon striking the target boundary at impact velocity v_{imp} , high-pressure planar shock waves propagate forward into the target and backward into the projectile casing. The shock wave velocity U_s and particle velocity U_p satisfy the linear shock Hugoniot equation of state (EOS):

$$U_s = C_0 + S \cdot U_p \quad (5)$$

At the impact interface, continuity of normal stress and particle velocity dictates:

$$U_p = \frac{v_{imp}}{1 + \sqrt{\frac{\rho_t}{\rho_p}}} \quad (6)$$

$$P_{shock} = \rho_p \cdot U_s \cdot U_p = \rho_p \cdot (C_{0,p} + S_p U_p) \cdot U_p \quad (7)$$

The explosive payload survivability is evaluated via the **Walker-Wasley shock initiation criterion**:

$$E_{shock} = P_{trans}^2 \cdot \tau \quad (8)$$

$$\text{State} = \begin{cases} \text{Survives (Intact)} & E_{shock} < E_c \\ \text{Premature Detonation} & E_{shock} \geq E_c \end{cases} \quad (9)$$

where $E_c \approx 3.0 \times 10^{15} \text{ Pa}^2 \cdot \text{s}$ is the critical initiation energy threshold.

C. Subterranean Deceleration: Two-Phase Forrestal Model

Ground penetration resistance is modeled using Forrestal's semi-analytical cavity expansion framework:

• Phase I: Surface Cratering ($z < 2D$):

$$F_z(z) = \frac{\pi D^2}{2} \cdot \sigma_s \cdot \left(\frac{z}{2D} \right) \quad (10)$$

• Phase II: Deep Tunneling ($z \geq 2D$):

$$F_z(v) = -\frac{\pi D^2}{4} (S \cdot f'_c(\dot{\epsilon}) + N \cdot \rho_t \cdot v^2) \quad (11)$$

where $S = 82.6 \cdot (f'_c)^{-0.544}$ is the target resistance factor, and $f'_c(\dot{\epsilon}) = f'_{c,static} \cdot \text{DIF}$ accounts for the CEB-FIP Dynamic Increase Factor.

D. Hydrodynamic Rod Erosion (Walker-Anderson Model / WAPM)

When instantaneous dynamic pressure $P_{dyn} = \frac{1}{2} \rho_t v^2$ exceeds casing dynamic flow stress Y_p , the penetrator undergoes hydrodynamic plastic erosion. Interface velocity u satisfies the modified Alekseevskii-Tate momentum balance:

$$Y_p + \frac{1}{2} \rho_p (v - u)^2 = R_t + \frac{1}{2} \rho_t u^2 \quad (12)$$

The rates of rod erosion $\frac{dL}{dt}$ and mass loss $\frac{dm}{dt}$ are:

$$\frac{dL}{dt} = -(v - u), \quad \frac{dm}{dt} = \rho_p \cdot A \cdot \frac{dL}{dt} \quad (13)$$

The theoretical hydrodynamic limit is $P_{hydro} = L_0 \sqrt{\frac{\rho_p}{\rho_t}}$.

III. COMPUTATIONAL ARCHITECTURE

The system unifies four synchronized subsystems:

- 1) **AI Orchestrator Layer ('src/AI')**: Built with Gemini 2.5 Flash via REST API. Provides automated hypothesis formulation ('researchConductor') and academic article synthesis ('articleWriter').
- 2) **Automation & Database Layer ('src/Automation')**: Express.js backend managing asynchronous execution, 30s watchdog hang protection, and chunked MongoDB streaming (1,000 frames/batch) with cryptographic session scoping.
- 3) **Native C++23 Physics Core ('src/simulation')**: High-speed numerical solver supporting headless '-json-input' direct configuration parsing via 'nlohmann::json'.
- 4) **Interactive 3D WebGL Engine ('3d_vizualizer.html')**: *Physics - boundreal - timeThree.jsrendererfeaturingPlanckblackbodythermalradiation*

IV. NUMERICAL EXPERIMENTS & RESULTS

TABLE I
GBU-57 CASING OPTIMIZATION AGAINST 70 MPA CONCRETE (6 AUTONOMOUS CYCLES)

Metric	Value	Std Dev (σ)
Mean Penetration Depth	8.22 m	$\pm 2.05 \text{ m}$
Maximum Recorded Depth	10.27 m	—
Minimum Recorded Depth	6.17 m	—
Average Impact Velocity	537.4 m/s (M 1.58)	$\pm 12.4 \text{ m/s}$
Mean Kinetic Energy	1.960 GJ	$\pm 0.08 \text{ GJ}$
Peak Shock Pressure (P_{shock})	3.620 GPa	$\pm 0.35 \text{ GPa}$
Casing Survivability Rate	100.0%	0.0% Failure
Dominant Regime	Rigid Tunneling	100.0%

A. Sequential Salvo Synergy

Simulations evaluating multi-bomb salvo drops (Operation Midnight Hammer configuration) demonstrated non-linear depth enhancement:

- **Bomb #1 (Lead Strike)**: Penetration depth = 8.22 m.
- **Bomb #2 (Follow-on)**: Cumulative depth = **17.48 m** (+12.7% over additive prediction).
- **Bomb #4 (Salvo Direct)**: Cumulative depth = **36.14 m** (+22.1% over additive prediction).

V. MACHINE LEARNING ROADMAP (V4.0)

The V4.0 roadmap incorporates:

- 1) **LibTorch Surrogate Physics:** Embedding TorchScript (‘.pt’) deep neural network surrogates inside C++ to achieve $O(1)$ sub-microsecond crater predictions ($> 1,000,000$ runs/sec).
- 2) **Reinforcement Learning Smart Fuzing:** An embedded deep RL policy evaluating instantaneous g -force and shock pressure to trigger subterranean detonation at maximum structural damage.

VI. CONCLUSION

The **MOP Simulator V3.0** demonstrates a scalable framework uniting high-performance C++ continuum mechanics with automated cloud-native web orchestration and generative AI scientific synthesis. The open-source platform provides a robust tool for researchers investigating high-strain-rate impact dynamics.

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